

One might ask whether a space dominated by a finite CW complex is homotopy equivalent to a finite CW complex. In the simply-connected case this follows from Proposition 4C.1 since such a space has finitely generated homology groups. But there are counterexamples in the general case; see [Wall 1965].

In view of Corollary A.10 the preceding proposition implies:

|| **Corollary A.12.** *A compact manifold is homotopy equivalent to a CW complex.* $\square$

One could ask more refined questions. For example, do all compact manifolds have CW complex structures, or even simplicial complex structures? Answers here are considerably harder to come by. Restricting attention to closed manifolds for simplicity, the present status of these questions is the following. For manifolds of dimensions less than 4, simplicial complex structures always exist. In dimension 4 there are closed manifolds that do not have simplicial complex structures, while the existence of CW structures is an open question. In dimensions greater than 4, CW structures always exist, but whether simplicial structures always exist is unknown, though it is known that there are n -manifolds not having simplicial structures locally isomorphic to any linear simplicial subdivision of $\mathbb{R}^n$, for all $n \geq 4$. For more on these questions, see [Kirby & Siebenmann 1977] and [Freedman & Quinn 1990].

Exercises

1. Show that a covering space of a CW complex is also a CW complex, with cells projecting homeomorphically onto cells.
2. Let X be a CW complex and x_0 any point of X . Construct a new CW complex structure on X having x_0 as a 0-cell, and having each of the original cells a union of the new cells. The latter condition is expressed by saying the new CW structure is a **subdivision** of the old one.
3. Show that a CW complex is path-connected iff its 1-skeleton is path-connected.
4. Show that a CW complex is locally compact iff each point has a neighborhood that meets only finitely many cells.
5. For a space X , show that the identity map $X_c \rightarrow X$ induces an isomorphism on π_1 , where X_c denotes X with the compactly generated topology.

The Compact-Open Topology

By definition, the compact-open topology on the space X^Y of maps $f: Y \rightarrow X$ has a subbasis consisting of the sets $M(K, U)$ of mappings taking a compact set $K \subset Y$ to an open set $U \subset X$. Thus a basis for X^Y consists of sets of maps taking a finite number of compact sets $K_i \subset Y$ to open sets $U_i \subset X$. If Y is compact, which is the only case we consider in this book, convergence to $f \in X^Y$ means, loosely speaking,

that finer and finer compact covers $\{K_i\}$ of Y are taken to smaller and smaller open covers $\{U_i\}$ of $f(Y)$. One of the main cases of interest in homotopy theory is when $Y = I$, so X^I is the space of paths in X . In this case one can check that a system of basic neighborhoods of a path $f: I \rightarrow X$ consists of the open sets $\bigcap_i M(K_i, U_i)$ where the K_i 's are a partition of I into nonoverlapping closed intervals and U_i is an open neighborhood of $f(K_i)$.

The compact-open topology is the same as the topology of uniform convergence in many cases:

Proposition A.13. *If X is a metric space and Y is compact, then the compact-open topology on X^Y is the same as the metric topology defined by the metric $d(f, g) = \sup_{y \in Y} d(f(y), g(y))$.*

Proof: First we show that every open ε -ball $B_\varepsilon(f)$ about $f \in X^Y$ contains a neighborhood of f in the compact-open topology. Since $f(Y)$ is compact, it is covered by finitely many balls $B_{\varepsilon/3}(f(y_i))$. Let $K_i \subset Y$ be the closure of $f^{-1}(B_{\varepsilon/3}(f(y_i)))$, so K_i is compact, $Y = \bigcup_i K_i$, and $f(K_i) \subset B_{\varepsilon/2}(f(y_i)) = U_i$, hence $f \in \bigcap_i M(K_i, U_i)$. To show that $\bigcap_i M(K_i, U_i) \subset B_\varepsilon(f)$, suppose that $g \in \bigcap_i M(K_i, U_i)$. For any $y \in Y$, say $y \in K_i$, we have $d(g(y), f(y_i)) < \varepsilon/2$ since $g(K_i) \subset U_i$. Likewise we have $d(f(y), f(y_i)) < \varepsilon/2$, so $d(f(y), g(y)) \leq d(f(y), f(y_i)) + d(g(y), f(y_i)) < \varepsilon$. Since y was arbitrary, this shows $g \in B_\varepsilon(f)$.

Conversely, we show that for each open set $M(K, U)$ and each $f \in M(K, U)$ there is a ball $B_\varepsilon(f) \subset M(K, U)$. Since $f(K)$ is compact, it has a distance $\varepsilon > 0$ from the complement of U . Then $d(f, g) < \varepsilon/2$ implies $g(K) \subset U$ since $g(K)$ is contained in an $\varepsilon/2$ -neighborhood of $f(K)$. So $B_{\varepsilon/2}(f) \subset M(K, U)$. $\square$

The next proposition contains the essential properties of the compact-open topology from the viewpoint of algebraic topology.

Proposition A.14. *If Y is locally compact, then:*

- (a) *The evaluation map $e: X^Y \times Y \rightarrow X$, $e(f, y) = f(y)$, is continuous.*
- (b) *A map $f: Y \times Z \rightarrow X$ is continuous iff the map $\hat{f}: Z \rightarrow X^Y$, $\hat{f}(z)(y) = f(y, z)$, is continuous.*

In particular, part (b) provides the point-set topology justifying the adjoint relation $\langle \Sigma X, Y \rangle = \langle X, \Omega Y \rangle$ in §4.3, since it implies that a map $\Sigma X \rightarrow Y$ is continuous iff the associated map $X \rightarrow \Omega Y$ is continuous, and similarly for homotopies of such maps. Namely, think of a basepoint-preserving map $\Sigma X \rightarrow Y$ as a map $f: I \times X \rightarrow Y$ taking $\partial I \times X \cup \{x_0\} \times I$ to the basepoint of Y , so the associated map $\hat{f}: X \rightarrow Y^I$ has image in the subspace $\Omega Y \subset Y^I$. A homotopy $f_t: \Sigma X \rightarrow Y$ gives a map $F: I \times X \times I \rightarrow Y$ taking $\partial I \times X \times I \cup I \times \{x_0\} \times I$ to the basepoint, with $\hat{F}$ a map $X \times I \rightarrow \Omega Y \subset Y^I$ defining a basepoint-preserving homotopy $\hat{f}_t$.

Proof: (a) For $(f, y) \in X^Y \times Y$ let $U \subset X$ be an open neighborhood of $f(y)$. Since Y is locally compact, continuity of f implies there is a compact neighborhood $K \subset Y$ of y such that $f(K) \subset U$. Then $M(K, U) \times K$ is a neighborhood of (f, y) in $X^Y \times Y$ taken to U by e , so e is continuous at (f, y) .

(b) Suppose $f: Y \times Z \rightarrow X$ is continuous. To show continuity of $\hat{f}$ it suffices to show that for a subbasic set $M(K, U) \subset X^Y$, the set $\hat{f}^{-1}(M(K, U)) = \{z \in Z \mid f(K, z) \subset U\}$ is open in Z . Let $z \in \hat{f}^{-1}(M(K, U))$. Since $f^{-1}(U)$ is an open neighborhood of the compact set $K \times \{z\}$, there exist open sets $V \subset Y$ and $W \subset Z$ whose product $V \times W$ satisfies $K \times \{z\} \subset V \times W \subset f^{-1}(U)$. So W is a neighborhood of z in $\hat{f}^{-1}(M(K, U))$. (The hypothesis that Y is locally compact is not needed here.)

For the converse half of (b) note that f is the composition $Y \times Z \rightarrow Y \times X^Y \rightarrow X$ of $\mathbb{1} \times \hat{f}$ and the evaluation map, so part (a) gives the result. $\square$

Proposition A.15. *If X is a compactly generated Hausdorff space and Y is locally compact, then the product topology on $X \times Y$ is compactly generated.*

Proof: First a preliminary observation: A function $f: X \times Y \rightarrow Z$ is continuous iff its restrictions $f|_{C \times Y}: C \times Y \rightarrow Z$ are continuous for all compact $C \subset X$. For, using (b) of the previous proposition, the first statement is equivalent to $\hat{f}: X \rightarrow Z^Y$ being continuous and the second statement is equivalent to $\hat{f}|_C: C \rightarrow Z^Y$ being continuous for all compact $C \subset X$. Since X is compactly generated, the latter two statements are equivalent.

To prove the proposition we just need to show the identity map $X \times Y \rightarrow (X \times Y)_c$ is continuous. By the previous paragraph, this is equivalent to continuity of the inclusion maps $C \times Y \rightarrow (X \times Y)_c$ for all compact $C \subset X$. Since Y is locally compact, it is compactly generated, and C is compact Hausdorff hence locally compact, so the same reasoning shows that continuity of $C \times Y \rightarrow (X \times Y)_c$ is equivalent to continuity of $C \times C' \rightarrow (X \times Y)_c$ for all compact $C' \subset Y$. But on the compact set $C \times C'$, the two topologies on $X \times Y$ agree, so we are done. (This proof is from [Dugundji 1966].) $\square$

Proposition A.16. *The map $X^{Y \times Z} \rightarrow (X^Y)^Z$, $f \mapsto \hat{f}$, is a homeomorphism if Y is locally compact Hausdorff and Z is Hausdorff.*

Proof: First we show that a subbasis for $X^{Y \times Z}$ is formed by the sets $M(A \times B, U)$ as A and B range over compact sets in Y and Z respectively and U ranges over open sets in X . Given a compact $K \subset Y \times Z$ and $f \in M(K, U)$, let K_Y and K_Z be the projections of K onto Y and Z . Then $K_Y \times K_Z$ is compact Hausdorff, hence normal, so for each point $k \in K$ there are compact sets $A_k \subset Y$ and $B_k \subset Z$ such that $A_k \times B_k$ is a compact neighborhood of k in $f^{-1}(U) \cap (K_Y \times K_Z)$. By compactness of K a finite number of the products $A_k \times B_k$ cover K . Discarding the others, we then have $f \in \bigcap_k M(A_k \times B_k, U) \subset M(K, U)$, which shows that the sets $M(A \times B, U)$ form a subbasis.

Under the bijection $X^{Y \times Z} \rightarrow (X^Y)^Z$ these sets $M(A \times B, U)$ correspond to the sets $M(B, M(A, U))$, so it will suffice to show the latter sets form a subbasis for $(X^Y)^Z$. We

show more generally that X^Y has as a subbasis the sets $M(K, V)$ as V ranges over a subbasis for X and K ranges over compact sets in Y , assuming that Y is Hausdorff.

Given $f \in M(K, U)$, write U as a union of basic sets U_α with each U_α an intersection of finitely many sets $V_{\alpha,j}$ of the given subbasis. The cover of K by the open sets $f^{-1}(U_\alpha)$ has a finite subcover, say by the open sets $f^{-1}(U_i)$. Since K is compact Hausdorff, hence normal, we can write K as a union of compact subsets K_i with $K_i \subset f^{-1}(U_i)$. Then f lies in $M(K_i, U_i) = M(K_i, \bigcap_j V_{ij}) = \bigcap_j M(K_i, V_{ij})$ for each i . Hence f lies in $\bigcap_{i,j} M(K_i, V_{ij}) = \bigcap_i M(K_i, U_i) \subset M(K, U)$. Since $\bigcap_{i,j} M(K_i, V_{ij})$ is a finite intersection, this shows that the sets $M(K, V)$ form a subbasis for $(X^Y)^Z$. $\square$

Proposition A.17. *If $f: X \rightarrow Y$ is a quotient map then so is $f \times \mathbb{1}: X \times Z \rightarrow Y \times Z$ whenever Z is locally compact.*

This can be applied when $Z = I$ to show that a homotopy defined on a quotient space is continuous.

Proof: Consider the diagram at the right, where W is $Y \times Z$ with the quotient topology from $X \times Z$, with g the quotient map and h the identity. Every open set in $Y \times Z$ is open in W since $f \times \mathbb{1}$ is continuous, so it will suffice to show that h is continuous.

$$\begin{array}{ccc} X \times Z & \xrightarrow{f \times \mathbb{1}} & Y \times Z \\ & \searrow g & \swarrow h \\ & W & \end{array}$$

Since g is continuous, so is the associated map $\hat{g}: X \rightarrow W^Z$, by Proposition A.14. This implies that $\hat{h}: Y \rightarrow W^Z$ is continuous since f is a quotient map. Applying Proposition A.14 again, we conclude that h is continuous. $\square$

Bibliography

Books

- J. F. ADAMS, *Algebraic Topology: a Student's Guide*, Cambridge Univ. Press, 1972.
- J. F. ADAMS, *Stable Homotopy and Generalised Homology*, Univ. of Chicago Press, 1974.
- J. F. ADAMS, *Infinite Loop Spaces*, Ann. of Math. Studies 90, 1978.
- A. ADEM and R. J. MILGRAM, *Cohomology of Finite Groups*, Springer-Verlag, 1994.
- M. AIGNER and G. ZIEGLER, *Proofs from THE BOOK*, Springer-Verlag, 1999.
- P. ALEXANDROFF and H. HOPF, *Topologie*, Chelsea, 1972 (reprint of original 1935 edition).
- M. A. ARMSTRONG, *Basic Topology*, Springer-Verlag, 1983.
- M. F. ATIYAH, *K-Theory*, W. A. Benjamin, 1967.
- H. J. BAUES, *Homotopy Type and Homology*, Oxford Univ. Press, 1996.
- D. J. BENSON, *Representations and Cohomology, Volume II: Cohomology of Groups and Modules*, Cambridge Univ. Press, 1992.
- D. J. BENSON, *Polynomial Invariants of Finite Groups*, Cambridge Univ. Press, 1993.
- R. BOTT and L. TU, *Differential Forms in Algebraic Topology*, Springer-Verlag GTM 82, 1982.
- G. BREDON, *Topology and Geometry*, Springer-Verlag GTM 139, 1993.
- K. BROWN, *Cohomology of Groups*, Springer-Verlag GTM 87, 1982.
- R. BROWN, *The Lefschetz Fixed Point Theorem*, Scott Foresman, 1971.
- M. COHEN, *A Course in Simple-Homotopy Theory*, Springer-Verlag GTM 10, 1973.
- J. DIEUDONNÉ, *A History of Algebraic and Differential Topology 1900-1960*, Birkhäuser, 1989.
- A. DOLD, *Lectures on Algebraic Topology*, Springer-Verlag, 1980.
- J. DUGUNDJI, *Topology*, Allyn & Bacon, 1966.
- H.-D. EBBINGHAUS et al., *Numbers*, Springer-Verlag GTM 123, 1991.
- S. EILENBERG and N. STEENROD, *Foundations of Algebraic Topology*, Princeton Univ. Press, 1952.
- Y. FÉLIX, S. HALPERIN, and J.-C. THOMAS, *Rational Homotopy Theory*, Springer-Verlag GTM 205, 2001.
- R. FENN, *Techniques of Geometric Topology*, Cambridge Univ. Press, 1983.
- A. T. FOMENKO and D. B. FUKS, *A Course in Homotopic Topology*, Izd. Nauka, 1989. (In Russian; an English translation of an earlier version was published by Akadémiai Kiadó, Budapest, 1986.)
- M. FREEDMAN and F. QUINN, *Topology of 4-Manifolds*, Princeton Univ. Press, 1990.